MAHATMA GANDHI UNIVERSITY, KOTTAYAM

Fifth Semester Regular Examination, October 2026

MG5DSEPHY304 – Physics of Advanced Materials | Maximum Marks: 50

Scheme of Valuation

PART A – Short-answer questions (2 marks each)-Answer any 10

1. Which are the different forms in which abrasives are used?

Ans: Abrasives are supplied as bonded abrasives, coated abrasives, and loose/free abrasive grains or powders.

Any two types-2 marks

Any one type-1 marks

2. Name the three typical types of stress–strain behaviour exhibited by polymeric materials.

Ans: Brittle polymer behaviour: fracture at low strain with little plastic deformation, Plastic behavior and highly elastic (elastomeric) behaviour.

All three types- 2 marks,

Any two type- 1 mark

Any one type- 0.5 mark

3. Give two examples of plastics and mention one important application of each.

Ans: Polyethylene (packaging films/containers); PVC (pipes/cable insulation); polystyrene (foam packaging); PTFE (chemical-resistant seals/non-stick coatings); PET (bottles).

Any two types with example: 2 marks.

Any one with example: 1 mark

4. Define electron mobility. State its SI unit.

Ans: Electron mobility is the magnitude of the drift velocity acquired by electrons per unit applied electric field: $\mu = |v_d|/E$; SI unit: $m^2\ V^{-1}\ s^{-1}$.

Definition-1 Mark

Unit- 1 mark

5. How do crazes propagate relative to tensile stress?

Ans: Crazes generally form and extend approximately perpendicular (normal) to the direction of the applied tensile stress; they contain microvoids bridged by oriented polymer fibrils.

Any two point-2 marks

Any one point-1 marks

6. Give two examples of structural clay products.

Ans: bricks, building tiles, sewer pipes, drainage pipes, roofing tiles, structural ceramic blocks.

Any two example-2 marks

Any one example -1 marks

7. What is meant by spontaneous polarization in a dielectric material?

Ans: A non-zero, permanent electric polarization that exists in a material even in the absence of an externally applied electric field; characteristic of ferroelectric materials below their Curie temperature.

Any two point-2 marks

Any one point-1 marks

8. State any two important properties of ultra-high-molecular-weight polyethylene (UHMWPE).

Ans: very high abrasion/wear resistance; high impact strength/toughness; low coefficient of friction; excellent chemical resistance; high molecular mass; good biocompatibility in suitable grades.

Any two properties-2 marks

Any one properties-1 marks

9. What is meant by the freeze-out region in an extrinsic semiconductor?

Ans: The low-temperature region in which thermal energy is insufficient to ionize most donor or acceptor dopants. Many carriers remain bound to impurity atoms, so the free-carrier concentration and conductivity are low.

Any two point-2 marks

Any one point-1 marks

10. Write a note about ferromagnetic materials.

Ans: Atomic magnetic moments align parallel in domains due to exchange interactions, producing spontaneous magnetization within domains; They show strong attraction to magnetic fields and typically exhibit hysteresis/remanence; above the Curie temperature, long-range ferromagnetic order is lost.

Any two point-2 marks

Any one point-1 marks

If domain structure is drawn and given appropriate explanation **– 2 marks**

Structure only**- 1 mark**

11. State the fundamental cause of thermal expansion in solids at the atomic/microscopic level.

Ans: Anharmonicity/asymmetry of the interatomic potential: as temperature increases, atoms vibrate with larger amplitude and their average separation increases. Purely harmonic vibrations would not produce ordinary thermal expansion.

Any two point-2 marks

Any one point-1 marks

12. Why are highly amorphous polymers transparent, whereas highly crystalline polymers of the same composition are translucent or opaque?

Ans: Amorphous polymers are structurally more uniform and have fewer refractive-index discontinuities to scatter visible light. Crystalline regions/spherulites and interfaces with amorphous regions scatter light because of refractive-index differences, making the material translucent or opaque.

Any two point-2 marks

Any one point-1 marks

PART B – Short-essay questions (6 marks each)-Answer any 5

13. Explain the two phases present in composites with example.

- **Ans:** Matrix phase: continuous phase that binds the reinforcement, transfers load, protects it from the environment and gives the component its shape. (**2 marks**)
- Dispersed/reinforcement phase: fibers, particles or flakes distributed in the matrix; improves strength, stiffness, wear resistance or other desired properties. (**2 marks**)
- Explain that the combination gives properties not obtained from either constituent alone. (**1 mark**)
- Correct example, e.g., glass-fibre-reinforced polymer (glass fibres in polymer matrix), carbon-fibre/epoxy, or concrete (aggregate in cementitious matrix). (**1 mark**)

14. Explain fracture toughness of ceramic materials. Write its expression and explain the terms involved.

- **Ans:** Fracture toughness measures a material's resistance to propagation of an existing crack; ceramics generally have low fracture toughness because they are brittle and have limited plastic deformation. (**1 mark**)
- Mode-I stress-intensity expression: $K_I = Y\ \sigma\ \sqrt{(\pi a)}$, with fracture occurring when K_I reaches K_{IC}. Alternatively, $K_{IC} = Y\ \sigma_c\ \sqrt{(\pi a)}$ at critical fracture. (**2 marks**)
- K_{IC}: mode-I stress intensity factor; K_{IC}: plane-strain fracture toughness (critical value); σ or σ_c: applied/critical tensile stress; a: characteristic crack length (state whether crack depth or half-length, according to geometry). (**1 mark**)
- Y: dimensionless geometry factor dependent on crack and specimen geometry/loading. (**1 mark**)
- Units of K_{IC}: MPa$\sqrt{m}$ (or Pa$\sqrt{m}$); correct explanation that larger K_{IC} means greater resistance to crack growth. (**1 mark**)

15. Explain the mechanism of electrical conduction in conjugated polymers.

- **Ans:** Conjugated polymers have alternating single and double bonds along the backbone, producing a delocalised π-electron system. (**1 mark**)
- Overlap of p orbitals creates π and π^* electronic states/bands; the undoped polymer commonly has a band gap and relatively low conductivity. (**1 mark**)
- Doping by oxidation (p-type) or reduction (n-type) introduces mobile charge carriers, commonly described as polarons/bipolarons or soliton-like defects depending on the polymer. (**2 marks**)
- Charge moves along the conjugated backbone through delocalised states and between chains by hopping/tunnelling; chain order and interchain contact influence conductivity. (**1 mark**)
- Example: doped polyaniline, polypyrrole or polyacetylene; doping can raise conductivity by many orders of magnitude. (**1 mark**)

16. Explain why electronic thermal conduction predominates in high-purity metals.

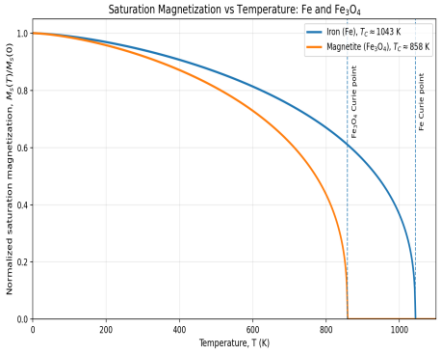
- **Ans:** Metals contain a large concentration of mobile conduction electrons. These electrons transport kinetic energy rapidly from hotter to colder regions. (**2 marks**)
- In high-purity metals, impurity and defect scattering of electrons is small, so electron mean free path and thermal conductivity are high; lattice/vibrational (phonon) conduction is usually a smaller contribution. (**2 marks**)
- The Wiedemann–Franz law relates electronic thermal conductivity κ_e to electrical conductivity σ: $\kappa_e/\sigma T \approx L$, where L is the Lorenz number. (**1 mark**)
- A correct comparison or explanation that electrons generally carry heat more effectively than phonons in metals. (**1 mark**)

17. Discuss the wide range of electrical conductivities exhibited by solid materials and explain how this range forms the basis for classifying materials electrically.

- **Ans:** Electrical conductivity spans many orders of magnitude, approximately from about 10^{-18} S m^{-1} for good insulators to 10^7–10^8 S m^{-1} for good conductors; values depend on material, temperature, purity and microstructure. (**2 marks**)
- Conductors/metals have many mobile electrons and high conductivity; semiconductors have intermediate, controllable conductivity depending on temperature, doping and illumination; insulators have very few mobile charge carriers and very low conductivity. (**2 marks**)
- Band explanation: metals have partially filled/overlapping bands, semiconductors have a relatively small band gap, and insulators have a large band gap. (**1 mark**)
- Give suitable examples: Cu/Al (conductors), Si/Ge (semiconductors), glass/porcelain/polymers (insulators). (**1 mark**)

18. Plot the saturation magnetization as a function of temperature for iron and Fe_3O_4.

- **Ans:** Axes correctly labelled: saturation magnetization M_s on vertical axis and temperature T on horizontal axis; magnetization decreases as temperature rises. (**2 marks**)
- Iron (Fe): M_s decreases with temperature and becomes zero at its Curie temperature, approximately 1043 K (770 °C). (**2 marks**)
- Fe_3O_4 (magnetite): M_s decreases with temperature and becomes zero at its Curie temperature, approximately 858 K (585 °C). Curves should be labelled and show Fe_3O_4 losing ferromagnetic/ferrimagnetic order at the lower temperature. (2 marks)
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19. Explain the increase of capacitance using the simplified model of polarisation within a dielectric material using a schematic diagram.

- **Ans:** For a parallel-plate capacitor without dielectric, $C_0 = \varepsilon_0 A/d$. (**1 mark**)
- When a dielectric is inserted, its molecules/ions polarise; bound charges appear on opposite surfaces and produce an internal field opposing the applied field, reducing the net field for a given free charge. (**2 marks**)
- More free charge can therefore be stored at the same applied voltage; capacitance increases. (**1 mark**)
- Expression: $C = \varepsilon A/d = \varepsilon_r \varepsilon_0 A/d = \varepsilon_r C_0$, where ε_r is relative permittivity (dielectric constant), A plate area and d separation. (**1 mark**)
- Clear labelled schematic showing plates, dielectric, polarisation/bound charges or induced dipoles, and field direction. (**1 mark**)

20. Explain any two optical properties of metals.

Ans: 1. High Reflectivity

- Metals exhibit high reflectivity because they contain a large number of free electrons. When electromagnetic radiation, such as visible light, falls on a metal surface, these electrons oscillate in response to the incident electric field and reradiate electromagnetic energy. A large fraction of the incident light is reflected from the surface.
- Reason: The interaction of incident light with free electrons in the metal.
- Examples: Silver, aluminum, and polished copper exhibit high reflectivity. (3marks)

2. Opacity (Low Transparency)

- Metals are generally opaque to visible light, meaning that light cannot pass through them in appreciable amounts. The free electrons interact strongly with incident electromagnetic radiation, causing reflection and absorption of the light. Any light that penetrates the metal is rapidly attenuated.
- - Reason: Strong interaction between electromagnetic radiation and free electrons, together with absorption within the metal.
- - Examples: Iron, copper, and aluminium sheets are opaque to visible light. (3 marks)

Reference

Callister, W. D. Jr., and Rethwisch, D. G., Materials Science and Engineering: An Introduction.